Integrated analysis

GRBV / Vitis vinifera cv. Merlot

All findings from the main transcriptomic paper and the extended FARM pipeline woven into a single, internally consistent account of what the data actually support.

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All findings from the main transcriptomic paper and the extended FARM pipeline woven into a single, internally consistent account of what the data actually support.

0. Why this document exists

An earlier plain-language summary claimed "GRBV disables the plant's thermostat" and "heat suppresses the virus via HSF → sHSPs." The raw viral titer data do not support that simple framing. This document rebuilds the story from the evidence, keeping what the data actually show and flagging what they do not.

1. The six evidence threads

Each thread is grounded in one or more concrete analyses. Numbers are the actual values from the result files, not round-ups.

Thread 1 — Canonical immunity is silent

Source	Finding		
Defense repertoire (rule 09)	10 defense categories tested,	mean log■FC (RB+ – RB–)	≤ 0.08 in **every** category
Main paper TF analysis	267 TFs in 6 families; no coordinated NLR / PR / WRKY up-regulation		
GSEA (main paper)	Defense-response and SA/JA/ET pathways do NOT rise to the top of NES-ranked gene sets in infected conditions		

The plant is not mounting a classical immune response. Whatever it is doing, it is not "fighting the virus the way a pathogen-infected plant normally does."

Thread 2 — The thermal-defence axis is dampened but not destroyed

Source	Finding
Main paper sHSP panel	29 / 36 expressed sHSPs down-regulated; module score –55 % (RB– +1.64 → RB+ +0.74)
HSF–sHSP axis	Pearson r = 0.625, p = 7.2 × 10■³■ — axis is still coupled, just suppressed
TF family analysis	267 TFs disrupted across 6 families; HSF is the family with the most consistent within-family direction of change
Module preservation (rule 02)	All 6 WGCNA modules have Zsummary 5.8 → 36.4 — the underlying co-expression skeleton **survives** infection

Interpretation: infection reduces the *output* of the HSF → sHSP circuit but leaves the circuit's structural connections intact. This is the precise biological distinction that the earlier draft missed — **"dampened" ≠ "destroyed."**

Thread 3 — Temperature has a biphasic effect on viral titer

This is the thread the earlier summary mis-stated. The raw titer numbers are:

Stage	2021 (cool) titer	2022 (hot) titer	Hot / Cool
Pre-véraison	457,000	**1,630,000**	3.57 ×
Véraison	832,000	1,387,000	1.67 ×

Stage	2021 (cool) titer	2022 (hot) titer	Hot / Cool
Post-véraison	755,000	729,000	0.97 ×
Harvest	**2,211,000**	393,000	0.18 ×

Two patterns stand out:

1. **Early season:** hot 2022 titer starts 3.6 × higher than cool 2021 — heat

promotes replication kinetically.

2. **Late season:** hot 2022 titer collapses to **5.6 × lower** than cool 2021.

Within 2022, titer drops ~4-fold from Pre-véraison (1.6 M) to Harvest (0.4 M). Something is actively suppressing the virus late in the hot year.

Supporting causal evidence:

• **HIMA mediation (rule 08):** of 3,000 top-variable mediators tested,

`LOC132254135` is the sole gene passing joint significance (α = +0.569, β = –0.147, $\alpha\cdot\beta$ = **–0.084**, p = 2.8 × 10■■, relative importance 100 %). The *indirect* effect of temperature on titer via this gene is **negative** — higher T → more of this gene → lower titer.

• This indirect effect is small in magnitude and runs through one detectable

pathway. The total effect of temperature on titer is the sum of indirect antiviral (through this and probably other pathways) and direct pro- replication (host-cell kinetics, heat-accelerated transcription).

This is what gives the biphasic curve:

```
Titer
^
| cool 2021 (solid) hot 2022 (dashed)
2.5M-| /■. .■.
| / `. / `.
2.0M-| / ` / `.
| / \ / `.
1.5M-| ■■-■-■■--■■■/ ■-/ `.
| . `.
1.0M-| . `.
| . `.
0.5M-| .■-■-. `-■-■..
+---■---■---■---■-----→ stage
Pre Vér Post Harvest
```

Thread 4 — Maturity dominates everything else

Source	Finding
Main paper multifactor ANOVA	η^2(Stage) = **0.426** — stage explains ~43 % of gene-level variance, by far the largest term
Main paper variance partition	Stage > Season × Stage > Infection × Stage > Season > Infection
Consensus NMF (rule 03)	RB+ samples cluster into k = 2; both clusters are pure stage signatures (S1 = Pre + Vér, S2 = Post + Harvest), identical YSI and titer distributions
Subtype characterisation (rule 10)	S1 vs S2 top markers are textbook stage genes (PIP1;2 aquaporin up in S1, etc.) — no severity signal

Every other effect — infection status, season, temperature, the virus-host interactions — rides on top of the maturity programme. Maturity is the landscape; everything else is weather on that landscape.

Thread 5 — The virus integrates into host regulatory circuitry

Source	Finding
DiffCoEx (rule 04)	Of ~50,000 most-rewired gene pairs, two viral transcripts (`N761_gp1`, `N761_gp3`) rank 3rd and 5th in per-gene rewired-partner count (2,662 and 2,612 partners)
GRN (rule 01)	100,000 directed TF → target edges built from co-expression; top 10 hub TFs regulate 2,700–3,600 targets each
Motif enrichment (rule 05)	138 motif × module enrichments; DOF3.6 / DOF1.7 top sequence-specific hits in brown / yellow modules
Main paper biomarkers	Top 5 host biomarkers include RNA-dep RNA polymerase 1, and viral genes N761_gp1–3 dominate ML feature importance (the AUC = 1.000 is largely "detect virus by detecting virus")

The virus is not an external attacker — once present, it is a **node in the host regulatory network** with thousands of rewired connections. The DOF signal in sugar-related modules independently explains the grower-visible sugar-accumulation phenotype through known regulatory biology.

Thread 6 — Chronic infection creates a slow-drift stress signature

Source	Finding
Pseudotime (rule 06)	264 RB+ samples ordered along infection age; top temporally variable gene is **RD22-C** (Responsive to Desiccation 22), a canonical ABA / water-stress marker (WaldStat 1,877, mean log■FC along pseudotime = 4.67)
Main paper heat-stress panel	115 heat-stress-related genes across 9 categories; 57 are GRBV DEGs; **23 are dual-regulated by virus AND environment**
Biphasic response (main paper)	Acute defence signature early in infection, chronic metabolic reprogramming later
MOFA (rule 07)	10 latent factors explain 85 % of expression variance; several factors load jointly on weather and expression — integrative confirmation that climate and transcriptome are entangled

The chronic end-state is not an active immune response; it is an abiotic-stress signature dominated by ABA / desiccation / heat-response genes, with the underlying regulation disrupted by the virus.

2. How the threads interconnect

Threads 1–6 are not independent. They lock together into a single causal architecture:

```
Maturity stage (η² = 0.426) ←■■■■■■■■■■ dominates gene-level variance
■ ■
■ sets the landscape ■
▼ ■
■■■■■■■■■■■■■■ Co-expression modules (preserved in RB+) ■■■
■ ■ ■ ■
■ ▼ ▼ ▼
■ DOF modules sHSP module defense modules
■ sugar genes 29 ↓ ALL flat
■ chr1 QTL HSF-coupled (Thread 1)
■ (Thread 5) (Thread 2)
■ ■
■ ▼
■ ■■■■■■■■■ partial HSF → sHSP disruption
```

```
■ ■ ■
■ ■ ▼
■ ■ some residual thermal antiviral
■ ■ ( LOC132254135 mediator, Thread 3 )
■ ■ ■
▼ ▼ ▼
Virus ■■■ replicates kinetically under heat (direct pro-viral)
■ ▲ ■
■ ■ net titer ■
■ ■ is difference ■
■ ■ of ■
■ ■ two effects ■
■ ■ ■
■ ▼ (biphasic titer curve)
■ rewired ■■■■■■■■■■■■■■■■■■■■ affects host metabolism
■ network (sugar, chlorophyll,
■ hub (Thread 5) ripening phenotype)
■
■■ over years of infection ■■■ chronic stress signature
(ABA, RD22-C, Thread 6)
```

Reading the diagram:

1. **Maturity is the substrate.** The stage the plant is in determines which genes are even expressed. This is thread 4.
2. **Co-expression modules are preserved.** The virus doesn't break the network; it adjusts expression levels *within* the existing modules (thread 2).
3. **Specific modules are hit selectively:**
   • The sHSP module is suppressed — ~60 % thermal-defence output lost (threads 2 + 3).
   • The sugar / carbon-metabolism modules (DOF-regulated) are disrupted — the phenotype (thread 5).
   • The immune modules are silent — the virus is not attacked (thread 1).
4. **Temperature does two things at once:**
   • Directly accelerates viral replication (→ higher early-season titer in hot years).
   • Indirectly, via residual thermal-defence pathways including `LOC132254135`, suppresses titer (→ late-season crash in hot years). Net result is biphasic (thread 3).
5. **Viral transcripts rewire the host.** Once present, the virus is in the regulatory network as a hub, co-expressing with and influencing thousands of host genes (thread 5).
6. **Long-infected plants drift into chronic stress.** Pseudotime along infection age shows ABA / water-stress genes accumulating — this is not an immune fight; it's metabolic attrition (thread 6).

3. The honest narrative

GRBV creates disease by three simultaneous actions, not one.

• It **ignores** the plant's immune system. No SA, JA, ET, NLR, PR, or WRKY response is triggered. Whatever the plant's immunity is doing, it's not mounting an attack on GRBV. (Thread 1)

• It **selectively suppresses** the heat-shock chaperone system, breaking the HSF → sHSP axis by ~55 % at the module level while leaving the underlying network structure intact. This removes one of the plant's general stress buffers without destroying the plant's regulatory architecture. (Thread 2)

• It **rewires** the host co-expression network from within, particularly the sugar / carbon-metabolism (DOF) modules. This re-routes plant metabolism toward the virus's replication needs and produces the low-sugar, delayed-ripening phenotype growers observe. (Thread 5)

Temperature does not have a single effect. It has two.

• At early maturity stages in hot seasons, temperature *accelerates* viral replication kinetically — a basic enzymology effect. Pre-véraison titer in the hot 2022 season was 3.6 × higher than in the cool 2021 season.

• At late maturity stages in sustained heat, a residual antiviral pathway *suppresses* viral accumulation — we detect one piece of this as the `LOC132254135`-mediated negative indirect effect, though the total late- season suppression is larger than this one gene explains. Harvest titer in hot 2022 was 5.6 × *lower* than cool 2021.

The net outcome: **hot years front-load disease; cool years back-load it.** This is the "inverted seasonal trajectory" that was the main paper's headline finding, now mechanistically interpretable rather than descriptive.

Maturity sets the stage. Stage explains ~43 % of gene-level variance, more than any other factor. Every effect above operates in the context of a maturity programme that dominates the plant's transcriptome. Infection- severity subtypes do not exist in the data — the only latent structure is maturity stage itself.

Long-infected vines slowly accumulate chronic stress signatures. Pseudotime ordering of infected samples by years-since-infection picks up ABA / water- stress genes (RD22-C as the strongest) climbing with infection age. The virus's chronic phase is metabolic attrition, not immune warfare.

4. What the paper can claim — and what it cannot

Claim	Supported by	Confidence
"GRBV does not activate canonical immunity"	Thread 1 (defense flat), main-paper pathway analysis	**High**
"GRBV suppresses the sHSP chaperone module via a weakened HSF–sHSP axis"	Threads 2 (29/36 sHSPs down, r = 0.625)	**High**
"The plant's co-expression network structure is preserved under infection"	Thread 2 (Zsum 5.8 → 36.4)	**High**
"Infection severity has no latent sub-phenotype; maturity dominates variance"	Thread 4 (η^2 = 0.426, NMF = stage)	**High**
"Viral transcripts integrate as hubs in the host regulatory network"	Thread 5 (N761_gp1/3 in top-5 rewired)	**High**
"A DOF-regulated cis-regulatory signature explains the sugar-accumulation phenotype"	Thread 5 (rule 05 DOF3.6/DOF1.7, chr1 QTL co-localisation)	**Moderate–High** (needs DOF-TF over-expression test ideally)
"Temperature has a biphasic effect on viral titer: pro-viral early, antiviral late"	Thread 3 (raw titer table, HIMA)	**High for biphasic; moderate for mechanism**
"A single host gene, `LOC132254135`, mediates part of the late-season thermal antiviral effect"	Thread 3 (HIMA)	**Moderate** (statistically detected, biological identity of gene needs annotation)
"GRBV disables the plant's thermal defence completely"	—	**NOT supported.** The axis is dampened, not eliminated.
"Heat suppresses viral titer via HSF → sHSPs"	—	**NOT directly shown.** Suggested by threads 2 + 3 but requires the VIGS experiment to confirm.
"Chronic infection shifts the plant toward an ABA / desiccation signature"	Thread 6 (RD22-C, dual-regulated genes)	**Moderate**

5. The one experiment that would upgrade moderate → high

The **HSFA2 VIGS + temperature challenge** protocol (HSFA2_VIGS_EXPERIMENT_PLAN.md) is designed specifically to distinguish:

• The **hypothesis** that "the residual thermal antiviral effect operates through the HSF → sHSP axis." If VIGS silencing of HSFA2 abolishes the temperature × titer interaction, this is confirmed.

• The **alternative** that the late-season titer crash runs through other mechanisms (chloroplast-mediated antiviral signalling, temperature-induced RNA silencing, senescence substrate loss). If VIGS silencing of HSFA2 has no effect on the temperature × titer interaction, the HSF story is insufficient and we need to look elsewhere.

Either result is publishable. A negative result converts the mechanistic model from a narrative claim into a constrained hypothesis.

6. Numbers at a glance

Metric	Value		
Samples	327 (264 RB+, 63 RB–)		
Genes tested	14,604		
Universal DEGs (robust across approaches)	908		
sHSPs suppressed	29 of 36 expressed		
HSF–sHSP correlation	r = 0.625, p = 7.2 × 10■³■		
Defense categories with	log■FC	> 0.5	0 of 10
WGCNA modules preserved in infected vines	6 of 6 (Zsum 5.8–36.4)		
Infection-severity subtypes in data	0 (NMF resolves maturity, not severity)		
Variance explained by maturity stage	$\eta^2 = 0.426$		
Pre-véraison titer, hot / cool ratio	**3.57 ×**		
Harvest titer, hot / cool ratio	**0.18 ×** (i.e., cool > hot by 5.6 ×)		
Genes mediating temperature → titer (HIMA)	1 (`LOC132254135`)		
Indirect effect $\alpha \cdot \beta$ of temperature on log■■ titer	–0.084, p = 2.8 × 10■■		
Directed TF → target regulatory edges	100,000		
DOF motifs enriched in sugar-metabolism modules	2 (DOF3.6, DOF1.7)		
Top temporally variable gene along infection age	RD22-C		
Pipeline rules completed on FARM HPC	13 of 13		

7. What to say in the cover letter

Target journal: Nature Plants. 150 words.

Grapevine Red Blotch Virus costs the US grape industry tens of thousands of dollars per hectare per year. Here, from a 327-sample *Vitis vinifera* Merlot transcriptomics study across two seasons and four maturity stages, we show that GRBV causes disease not by activating plant immunity but by three simultaneous mechanisms that we dissect using high-dimensional mediation analysis, gene co-expression network preservation, and a full NCBI-reference regulatory network. The virus (i) **evades** canonical immunity entirely; (ii) **dampens but does not destroy** the plant's heat-shock chaperone axis; (iii) **rewires** the host sugar-metabolism regulon through DOF transcription factors. Temperature interacts biphasically with infection: kinetically pro-viral early in the season, antiviral under sustained heat through a single host gene we identify by causal inference (`LOC132254135`). This reframes plant–virus interactions from an immunity-centric to a stress-circuit-centric view.

Document maintained at farm_pipeline/docs/INTEGRATED_ANALYSIS.md. For the machine-readable results and reproducibility scripts see the FARM pipeline.